

Spectral Solver Design — Liouville / Sturm–Liouville / Chebyshev Methods for the Stratified Pressure Equation

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0. About this document

`stellar2d` 2026-05-01..03 `(anelastic_SL_spectral_design.md, liouville_SL_spectral_derivation.md, reduced_pressure_liouville.md, liouville_singulariry_causality.md)` repo

□□□□□□□□: □□□□□ docs/spectral_stratified_poisson_report_2026-05-03.md
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□□□□□□□: - docs/spectral_stratified_poisson_report_2026-05-03.md — □□□□□□□
 - docs/spectral_experiments.md — □□□□□□□ (Phase 0 / reduced-pressure / g-mode / polytropic
 index) - docs/singular_basis_survey_2026-05-02.md — GYRE / Dedalus □□

Part I Motivation and Problem Setup

1. Motivation

□□ **pseudo_spectral** □□□□□□□□ 2D □□□ NS ($\rho = \text{const}$, $\nabla \cdot \mathbf{\Gamma} = 0$), □□□□, □□-□□□□□. □□□□□□□□ $\rho_0(y)$, □□□□□ Poisson □□□□ □□□□□□□, Fourier □□□□□□□□, □□□□□ $\mathcal{O}(N \log N)$ □□□□.

□ □ □ □ □ □ □ □ □ □ :

1. **Liouville-SL** ☐ **(Part II-III):** ☐ $\hat{p} = \sqrt{\rho_0} q$ ☐ ☐ ☐ ☐ Poisson ☐ ☐ Schrödinger ☐ ☐ ☐ ☐ ☐ k_x ☐ ☐ Poisson ☐ ☐ “cuFFT + GEMM + ☐ ☐ + GEMM + cuFFT” ☐.
2. ☐ **Chebyshev collocation (Part IV):** Phase 0 ext+ ☐ ☐ ☐ ☐, ☐ Eddington ☐ ☐ ☐ ☐ ($n = 3, \sigma = 3$), raw Chebyshev ☐ ☐ ☐ ☐ ☐ ☐, ☐ ☐ SL ☐ ☐. ☐ **Phase 1** ☐ ☐ ☐.

□□□□□□□□ Chebyshev □□□□□□, □□□□□□□□□□. SL □□□□□ Phase 2+ □□□□□□.

2. Problem formulation

2.1

ρ_0 is a positive function, ρ_0

$$\nabla \cdot \left(\frac{1}{\rho_0(y)} \nabla p \right) = f(\mathbf{\Gamma}) \quad (2.1)$$

$\rho_0(y)$ is a function of y . ρ_0 is anelastic $\nabla \cdot (\rho_0 \mathbf{\Gamma}) = 0$, ρ_0 is a function of y ; ρ_0 is a function of y .

2.2 Fourier-in- x

x is a function of y , ρ_0 is a function of y . $p(x, y) = \sum_{k_x} \hat{p}(k_x, y) e^{ik_x x}$, k_x is a function of y . $1D$ is a function of y . ODE :

$$\frac{d}{dy} \left[\frac{1}{\rho_0(y)} \frac{d\hat{p}}{dy} \right] - \frac{k_x^2}{\rho_0(y)} \hat{p} = \hat{f}(k_x, y). \quad (2.2)$$

k_x^2 is a function of y . $1/\rho_0(y)$ is a function of y . SL is a function of y . $\frac{d}{dy} [(1/\rho_0) \frac{d}{dy}]$ is a function of y . k_x^2 is a function of y .

Part II-III (Liouville-SL) is a function of y . Part IV (Chebyshev collocation) is a function of y .

Part II Liouville Normal-Form Reduction

3. The Liouville substitution

3.1

$\hat{p}(y) = \sqrt{\rho_0(y)} q(y)$

$$\hat{p}(y) = \sqrt{\rho_0(y)} q(y), \quad (3.1)$$

ρ_0 is a function of y .

Proposition 1 (Liouville reduction). ρ_0 is a function of y .

$$\frac{d}{dy} \left[\frac{1}{\rho_0} \frac{d\hat{p}}{dy} \right] = \frac{1}{\sqrt{\rho_0}} [q'' + W(y) q],$$

ρ_0 is a function of y .

$$W(y) = \frac{\rho_0''}{2\rho_0} - \frac{3(\rho_0')^2}{4\rho_0^2}.$$

Proof. ρ_0 is a function of y .

$$\begin{aligned} \hat{p}' &= \frac{\rho_0'}{2\sqrt{\rho_0}} q + \sqrt{\rho_0} q', \\ \frac{1}{\rho_0} \hat{p}' &= \frac{\rho_0'}{2\rho_0^{3/2}} q + \frac{q'}{\sqrt{\rho_0}}. \end{aligned}$$

□□□□:

$$\begin{aligned}\frac{d}{dy} \left[\frac{1}{\rho_0} \hat{p}' \right] &= I_1 + I_2, \\ I_1 &= \left[\frac{\rho_0''}{2 \rho_0^{3/2}} - \frac{3(\rho_0')^2}{4 \rho_0^{5/2}} \right] q + \frac{\rho_0'}{2 \rho_0^{3/2}} q', \\ I_2 &= -\frac{\rho_0'}{2 \rho_0^{3/2}} q' + \frac{q''}{\sqrt{\rho_0}}.\end{aligned}$$

q' □□□, □

$$I_1 + I_2 = \frac{1}{\sqrt{\rho_0}} [q'' + W(y) q]. \quad \square$$

□

3.2 □□□□□□

□□ (2.2) □□□ $\sqrt{\rho_0}$:

$$\boxed{q'' + W(y) q - k_x^2 q = g(y), \quad g \equiv \sqrt{\rho_0} \hat{f}.} \quad (3.4)$$

□□ **Liouville–Schrödinger** □□

$$\mathcal{T} \equiv \frac{d^2}{dy^2} + W(y), \quad (3.5)$$

□□ (3.4) □□

$$[\mathcal{T} - k_x^2] q = g. \quad (3.6)$$

□□: $\mathcal{T}$ □□□ k_x . □□□□□□□□□□□□□□□□. □□□□□□□□□□□□□□ k_x □□□□□.

4. Spectral diagonalisation

4.1 □□□□

□□ Sturm-Liouville (□□□, □□□□ Schrödinger) □□□□□

$$\mathcal{T} \psi_n = \psi_n'' + W(y) \psi_n = -\mu_n \psi_n, \quad n = 0, 1, 2, \dots \quad (4.1)$$

□□□□□□□ (□□ / Dirichlet / Neumann) □, SL □□□□□:

1. □□□ $\{\mu_n\}$ □□□□□□□: $\mu_0 \leq \mu_1 \leq \mu_2 \leq \dots$
2. □□□□ $\{\psi_n\}$ □ $L^2([0, L_y])$ □□□□□□□□
3. □ $W \equiv 0$ (□□□□), ψ_n □□□□□ Fourier □, $\mu_n = n^2 \pi^2 / L_y^2$

Theorem 1 (□□□□□). $\{\psi_n\}$ □□□ k_x □□□□□ $\mathcal{T} - k_x^2$:

$$[\mathcal{T} - k_x^2] \psi_n = -(\mu_n + k_x^2) \psi_n.$$

4.2 $\square\square\square\square\square\square$

$\square\square q = \sum_n a_n \psi_n, g = \sum_n g_n \psi_n, \square\square (3.6)$:

$$a_n(k_x) = -\frac{g_n(k_x)}{\mu_n + k_x^2}. \quad (4.3)$$

$\square\square\square\square. \square\square\square 1/\rho_0 \square\square\square\square\square\square\square\square \{(\mu_n, \psi_n)\} \square. \square\square (k_x, n) \square\square\square\square\square\square\square\square\square, \square\square\square\square\square\square\square\square\square$
Fourier-Poisson $\square \hat{p}(\Gamma) = -\hat{f}(\Gamma)/|\Gamma|^2$.

5. Complete SL-GEMM algorithm

5.1 $\square\square\square (\square\square\square, \rho_0 \square\square\square\square\square)$

1. $\square \rho_0(y) \square\square W(y) \square (3.3)$
2. $\square$ 1D Schrödinger $\square\square\square\square (4.1), \square \{\mu_n, \psi_n\}$
3. $\square\square\square\square\square\square \Psi_{in} = \psi_n(y_i), \Psi \in \mathbb{R}^{N_y \times N_y}$
4. $\square\square\square \sqrt{\rho_0(y_i)} \square\square$

5.2 $\square\square\square\square$

Step	Operation	Formula	Cost
1	FFT in x	$f(x, y) \rightarrow \hat{f}(k_x, y)$	$\mathcal{O}(N_x N_y \log N_x)$
2	Weight	$g = \sqrt{\rho_0} \cdot \hat{f}$	$\mathcal{O}(N_x N_y)$
3	Fwd SL transform	$G = \Psi^\top g$	$\mathcal{O}(N_y^2 N_x)$
4	Pointwise divide	$Q_n(k_x) = -G_n/(\mu_n + k_x^2)$	$\mathcal{O}(N_x N_y)$
5	Inv SL transform	$q = \Psi Q$	$\mathcal{O}(N_y^2 N_x)$
6	Weight	$\hat{p} = \sqrt{\rho_0} \cdot q$	$\mathcal{O}(N_x N_y)$
7	IFFT in x	$\hat{p}(k_x, y) \rightarrow p(x, y)$	$\mathcal{O}(N_x N_y \log N_x)$

$\square\square\square\square: \square\square 3, 5 \square\square\square\square\square\square\square, \square\square\square\square \mathcal{O}(N_y^2 N_x) \square\square$ Poisson $\square$.

5.3 $\square\square\square\square\square$

$\square\square$	$y \square\square$	$\square (N \times N)$	GPU $\square\square$
Fourier ($\rho_0 = \text{const}$)	$\square\square$	$N^2 \log N$	cuFFT ($\square\square$)
Chebyshev + $\square\square\square$	Thomas	N^2	$\square\square\square\square$
SL-GEMM	GEMM	Γ^3	cuBLAS batched (tensor cores)
Iterative (CG/MG)	SpMV $\times k$	$N^2 k$	$\square\square\square\square$

$\square\square$ SL-GEMM $\square\square\square\square\square\square\square, \square\square\square\square\square\square\square$ GPU $\square\square\square\square\square\square \square\square\square\square (\square\square\square\square\square, \square\square\square\square, \square\square$ tensor cores). $\square N_y \lesssim 4096, \square\square$ DGEMM $\square$ wall-clock $\square\square\square$ latency-bound Thomas $\square\square\square\square\square\square\square\square$.

□□□□□□: □□□□□□□□□□□□, SL □□□□□□.

Scope □□ (2026-05-03 □□): □□□□□ 7× □□□□ $\sigma = 3/2$ □□□. □□□□□□□ $\sigma = 3$ (Eddington, ρ_0 □□□□), raw Chebyshev □□□□□□□, □□□□□□ operational □□. □ Part IV.

Part IV Direct Chebyshev Collocation (Phase 1 □□□)

8. □□ vs □□ σ □□□□□□

Phase 0 ext+ (2026-05-03) □□□□□: □□□ reduced-pressure □□ (2.2 □ 6.1) □ Chebyshev collocation, □□□□□□, □□□□□□ □□□□ σ □□□□□□□□.

8.1 □□□□

□ manufactured-solution Poisson □□ $\rho_0(r) = (1-r)^\sigma$, $\pi_{\text{exact}} = \sin(2\pi r)$, $f = [\rho_0 \pi']' - k_x^2 \rho_0 \pi$ (SymPy □□□□).

\begin{center}

N	$\sigma = 3$ □□	$\sigma = 3/2$ □□ (raw)
16	8.5×10^{-8}	2.7×10^{-3}
32	$\sim 10^{-10}$	5.1×10^{-4}
64	6.7×10^{-11}	1.8×10^{-4}
128	$\sim 10^{-9}$ (roundoff)	5.2×10^{-5}
256	3.2×10^{-9}	1.1×10^{-5}

\end{center}>

- $\sigma = 3$: □□□□□□□□□□ ($\sim 10^{-10}$ at $N = 64$), □□ roundoff □□□□ — □□ Chebyshev □□□□□□□□□□□□.
- $\sigma = 3/2$: □□ N^{-2} □□□□, □□ r^α □□□□□□.

8.2 □□□□□ (Trefethen Thm 7.2)

Chebyshev □□□□□□□□□□□□□□□□□□. □ $\rho_0(r) = (1-r)^\sigma$:

- $\sigma \in \mathbb{Z}_{\geq 0}$: ρ_0 □□□□, Chebyshev □□ □□□ ($\sigma+1$ □), □□□□□□□□□□□□. □□□□□□□ □□□□.
- $\sigma \notin \mathbb{Z}$: $(1-r)^\sigma$ □□□□□□□□□□, Chebyshev □□□□ $N^{-\sigma-1/2}$ □□. L_N □□ L □□□ □□□□, □□□□□ $N^{-\sigma-1/2}$ □□□□.

8.3 □□□□□□□□□□

Lane-Emden □□□□□□ $\theta \sim (\xi_1 - \xi)$, $\rho_0 \propto \theta^n$ □□ $\rho_0 \propto (R-r)^n$. □□□□ n □□□□□□□ σ .

\begin{center}

10.2 Performance

“Unified SL basis simultaneously diagonalises Poisson and g-mode operators” [10]. [11]:

- Chebyshev [12] (CGL [13])
- Dense linear algebra [14] (cuBLAS GEMM, VRAM [15]) [16]
- Poisson [17] k_x [18] LU [19] ([20])
- g-mode [21] **EVP**, [22]

[23], [24]. [25] Phase 0 ext+ (E5/E6) [26].

Part VI Liouville Singularity — Physical Interpretation

11. [27]: QM vs [28]

Liouville [29] $\rho_0 \rightarrow 0$ [30] $1/t^2$ [31], [32] Coulomb/centrifugal barrier [33], [34].

11.1 QM [35]: [36]

$$V(r) \text{ ([37])} \rightarrow \psi(r) \rightarrow 0 \text{ ([38])}$$

Coulomb [39] centrifugal barrier [40], $|\psi|^2$ [41] [42]”[43]”. Coulomb-Sturmian [44]Laguerre DVR[45] R-matrix matching [46] QM [47]”[48]”[49].

11.2 [50]: [51]

$$\rho_0(y) \rightarrow 0 \text{ ([52])} \rightarrow W(y) \rightarrow -\infty \text{ ([53])}$$

[54] $\rho_0(y)$ [55] (Lane-Emden, MESA). [56] [57] — [58]”. [59] $\nabla \cdot (\rho_0^{-1} \nabla p)$ [60] (degenerate elliptic), [61], [62]. W [63] $\sqrt{\rho_0}$ [64], [65], [66].

11.3 [67]

1. **QM** [68]. Coulomb-Sturmian [69]”[70]”, [71]”[72]”[73].
2. **Path A** [74]. [75] r^α [76] W [77], [78] QM [79] Coulomb-Sturmian [80] $r^\ell e^{-\alpha r}$, [81], [82]”[83]”[84].
3. [85]. Reduced-pressure (Part III) [86] [87] “[88]” [89] “[90]”, [91]; raw Chebyshev (Part IV) [92] ρ_0 [93], [94] σ [95], [96]”[97]”[98].

[99]: [100] QM [101] Coulomb [102], [103] [104] Poisson [105]. [106] $\sigma \in \mathbb{Z}$ [107]; [108] $\sigma \notin \mathbb{Z}$ [109] Jacobi [110] [111] (Dedalus) [112].

Part VII Phase Roadmap

12. Current status and forward plan

12.1 (Phase 0 ext+, 2026-05-02..03)

- Lane-Emden $n = 3/2$ $n = 3$
- Chebyshev $N = 48$ (192 DOF) GYRE full-gravity 4-var benchmark, $\max_{\text{rd}} 1.5 \times 10^{-6}$,
FD $N_r = 1024$ (4096 DOF) $21 \times \text{DOF} / 350 \times$
- (manufactured Poisson, Dirichlet Laplacian) $(10^{-13}-10^{-15})$
- Barycentric Lagrange “N $\neq$ N”
- Path A/B/C: Path A (raw Chebyshev) $n = 3$

docs/spectral_experiments.md docs/spectral_stratified_poisson_report_2026-05-03.md.

12.2 Phase 1: 2D Fourier-Chebyshev Boussinesq

- **x**: Fourier, pseudo_spectral cuFFT
- **y**: Chebyshev collocation, GPU cuBLAS dense GEMM $D^{(2)}$
- 2D Boussinesq + buoyancy. Poisson via Chebyshev + dense solve
- **Benchmark**: Rayleigh-Bénard Nu-Ra scaling (Ahlers)
- Gaussian $\rightarrow$ Eddington $n = 3$ polytrope

docs/phase1_2d_spectral_design.md.

12.3 Phase 2: Anelastic

- Boussinesq anelastic: $\nabla \cdot (\rho_0 \mathbf{\Gamma}) = 0$
- Chebyshev Poisson (raw reduced-pressure)
- **SL-GEMM**, dense solve GPU

12.4 Phase 3: ()

- EVP g-mode / p-mode
- runtime diagnostic
- **novelty**: 2D DNS
–

12.5 Phase 4+: ()

docs/sph_spectral_roadmap.md.

12.6 Bibliography

- JCP: “Sturm-Liouville spectral methods for stratified astrophysical flows — convergence regimes and GPU implementation”
- A&C / ApJS: “GPU anelastic pseudo-spectral with live eigenmode projection for convection-pulsation coupling diagnostics”

□□□□ (□ novelty); □□□□□□□□.

Appendix A GPU Implementation Considerations

A.1 Batched GEMM □□

□□ 3, 5 □□□□□□□□□□□□:

$$\mathbf{\Gamma} = \Psi^\top \mathbf{\Gamma}, \quad \mathbf{\Gamma} = \Psi \mathbf{\Gamma},$$

□□ $\mathbf{\Gamma}, \mathbf{\Gamma}, \mathbf{\Gamma}, \mathbf{\Gamma} \in \mathbb{R}^{N_y \times N_x}$. □□ GPU (NVIDIA Ampere/Hopper) FP64 GEMM □□ 1 TFLOP/s, □□□□ $\mathcal{O}(N)$ flops/byte □□ $N_y \geq 256$ □ compute-bound.

A.2 □□□□□

SL-GEMM Poisson solver □□□□ pseudo-spectral □□□□□ (□□ IFRK3 + cuFFT), □□□□□ $\hat{p} = -\hat{f}/|\mathbf{\Gamma}|^2$ □□□:

cuFFT (R2C in x) $\rightarrow$ cuBLAS DGEMM $\rightarrow$ □□□ $\rightarrow$ cuBLAS DGEMM $\rightarrow$ cuFFT (C2R in x).

□□□□□□□□ — IFRK3, skew-symmetric □□, 2/3 dealias, VRAM frame buffer — □□□□□□□□.

A.3 □□□□□

□□□□ $\Psi \in N_y^2$ doubles = 32 MiB at $N_y = 2048$. □□ □□□□ (100 MiB each at 2048^2) □ VRAM frame buffer (~10 GiB) □□.

Appendix B □□□□□□□□□

B.1 □□□□□□□□□

□□	□□□□	□□□□	□□□□□□
□□□□	ASH, Rayleigh	□□(□□) + Chebyshev/FD(□□) + □□	CPU □ GEMM □
GPU □□	hit3d, spectralDNS	Fourier □□□	□□□□□□
□□□ GPU CFD	□□ LES	FV + multigrid	□□□□□

□□	□□□□	□□□□	□□□□□□
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□□□□□□□□□□ “Liouville + SL + GPU GEMM + anelastic” □□□.

B.2 SL □□□□

- **Liouville normal form**: Sturm 1836; Liouville 1837; □ Zettl (2005) □□□□
- **SL □□□□□□□ PDE**: Boyd (2001) □□□□□□□□□□□□
- **Chebyshev □□□□□**: Trefethen (2013) Ch. 7 (□□ vs □□ □□□□□□□□□□)
- **Berrut-Trefethen 2004**: barycentric Lagrange, □□□□□□□ □□□□□□□□□□□□□□□□□

B.3 □ GYRE / Dedalus □□□

□□ docs/singular_basis_survey_2026-05-02.md. □□□□ **novelty** □□□□□□□□ “GPU 2D anelastic DNS + □□□□□□□”, □□ GYRE (1D □□□□□□□) □ Dedalus (□□□□ PDE □□) □ 1D □□ benchmark □□□.

Appendix C Historical record — 4 □□□□□□□□□□

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Part II §3-4	liouville_SL_spectral_derivation.md	§3-4
Part II §5	liouville_SL_spectral_derivation.md	§5 + anelastic_SL_spectral_design.md
Part III §6-7	reduced_pressure_liouville.md	§5
Part IV §8-9	polytropic_index_spectral_convergence_2026-05-03.md + □□□□	§3
Part V §10	□□ (Phase 0 ext+ E5/E6 □□)	□□ (Phase 0 ext+ E5/E6 □□)
Part VI §11	liouville_singularity_causality.md	□□
Part VII §12	□□ (Phase 0 ext+ □□□□)	□□ (Phase 0 ext+ □□□□)
Appendix A	anelastic_SL_spectral_design.md	§8 + liouville_SL_spectral_derivation.md
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Appendix B	anelastic_SL_spectral_design.md §7 + liou- ville_SL_spectral_derivation.md §9	

□□□□□□□□□□ **repo** □□□□□□□□, □□□□□□ **Update** □□□□□□□□□□. □□□□□□ + □□□□□□.

0. About this document

stellar2d Phase 0 Phase 0 ext+ , 2026-05-02..03
setup / / . repo :

- docs/anelastic_sl_phase0_2026-05-02.md (§1-2: SL Poisson)
- docs/reduced_pressure_experiments_2026-05-02.md (§3: reduced-pressure)
- docs/gmode_experiments_2026-05-02.md (§4-6: g-mode A K)
- docs/polytropic_index_spectral_convergence_2026-05-03.md (§7: σ)

: EXPECTED script , python scripts/<name>.py --verify
exit zero. EXPECTED commit .

-docs/spectral_solver_design.md— -docs/spectral_stratified_poisson_report.
— -docs/singular_basis_survey_2026-05-02.md — GYRE / Dedalus

Part I SL Poisson (Phase 0, 2026-05-02)

1. Lane-Emden Liouville W

1.1 Setup

Emden $\theta'' + (2/\xi)\theta' + \theta^n = 0$ ξ_1 .

1.2 Lane-Emden $n = 3/2$

ξ_1 (polytrope)	3.653754
ρ/ρ_c	$[1.0, 2 \times 10^{-5}]$
$W(y)$	$[-1.35 \times 10^6, -3.3]$
$\rho < 0.01$	$ W _{\max} = 1.35 \times 10^6$ ()
$r/R_* \in [0, 0.94]$ ($\rho > 0.01$)	$W \in [-398, -3.3]$ bounded

Script: scripts/anelastic_sl_phase0.py.

1.3

(). $n = 3/2$ $r/R_* \leq 0.94$ (,).
ASH/Rayleigh .

2026-05-03 : $n = 3/2$. $n = 3$ (Eddington), $\rho_0 \propto (R - r)^3$, Chebyshev
 . Part IV.

2. SL + Fourier

2.1

Interior FD (nodes $1..N-2$, Dirichlet), $N = 512$. `scipy.sparse.linalg.eigsh` 256.

2.2 Fourier (W=0)

$W(y) = 0$, SL $(n\pi/L)^2$.

n	μ_{SL}	$(n\pi/L)^2$	rel err
1	11.146	11.146	3.1×10^{-6}
20	4452.8	4458.4	1.26×10^{-3}

$n = 20$ rel err FD stencil $n^4\pi^4\Delta y^2/(12L^4) \approx 1.26 \times 10^{-3}$. $\square = \square \rightarrow \square$.

2.3 SL-Poisson manufactured-solution

N_{modes}	err_{L^2}	
5	3.00×10^{-2}	—
10	7.34×10^{-3}	4.1×
20	1.37×10^{-3}	5.4×
40	2.18×10^{-4}	6.3×
80	3.27×10^{-5}	6.7×
160	5.99×10^{-6}	5.5×
256	$\times 3.74010^{-6}$	$\square \square \square$ FD $\square$

$\square \square \square$: $\log(\text{err})$ vs $\log(N) \approx -3.5$, $\square \square \square$, $\square \square \square$. $\square \square \square \rho \rightarrow 0$, $\widetilde{W}$ $\square \square \square \square \square$, $\square \square \square \square \square$.

2.4 Sturm + Tassoul (Phase 0 ext, E1-E2)

- **E1 (Sturm):** ψ_n n $(\square \square 21/21 \square \square)$. $\square \square \square \square \square \square \square \square + \square \square \square \square$.
- **E2 (Tassoul):** ΔP_n $\square \square \square \square$, std/mean = 8×10^{-4} . $\square \square \square \square$ Tassoul 1980, $\square \square \square 2.9 \times \square \square \square \square$ Cowling slab $\square \square (\ell(\ell+1)/r^2)$.

2.5 vs cutoff (E3) — vs

Lane-Emden $\rho_{\text{threshold}}$:

$\begin{center}$

cutoff	r_{hi}	err(256)	slope
0.1	0.77	4.3×10^{-7}	-2.42
0.01	0.94	3.7×10^{-6}	-2.39
0.001	0.99	2.7×10^{-5}	-2.26
0.0001	0.997	1.5×10^{-4}	-1.78

\end{center}>

□□□ $\rho = 0$, □□□□ — □□□□□□□.

Gaussian-capped □□ $\rho(y) = \exp(-2y^2) + 0.05$ (□□□): err(256) = 8.3×10^{-7} , semilog slope = -0.049
→ **err** $\sim \exp(-0.05 \cdot N)$, □□□□.

□□□□□: > “The SL method is exponentially convergent for smooth stratification. > Algebraic convergence observed on Lane-Emden polytropes is entirely > attributable to the surface singularity $\rho(R_\star) = 0$, quantifiable > via cutoff scaling analysis.”

2.6 Brunt-Väisälä $N^2(\mathbf{r})$ vs Liouville $W(\mathbf{r})$ □□□□ (E4)

\begin{center}

□	□□	□□
$W(r)$	□□□□□ (ρ'')	SL Poisson inversion □ Liouville □
$N^2(r)$	□□ + □□ (Schwarzschild)	g-mode □□□□ / □□□□□

\end{center}>

□□: Lane-Emden □ γ -□□ polytrope, Schwarzschild □□, $N^2 \equiv 0$. □□□□□ ($\delta = 0.1 \sin 2\pi r$ □□□ T
□) □ $|N^2| \sim 1$, □ $|W| \sim 10\text{-}400$ □□.

2.7 Phase 0 □□□□□

\begin{center}

□□□	□□	□□
Lane-Emden $W(y)$ □□□□□	□	□□□□ cutoff □□
SL □□□ Fourier □□	□	= FD □□□□
□ 256 □□□□□□□□	□	scipy eigsh, <1s
SL-Poisson $\text{err}_{L^2} = 3.7 \times 10^{-6}$	□	□□□□
Sturm oscillation (E1)	□	21/21 □□□□
Tassoul asymptotic ΔP (E2)	□	□□□□□□
Exponential vs algebraic (E3)	□	□□□□□□□□
$N^2 \leftrightarrow W$ □□□□ (E4)	□	Phase 2/3 scope □□

\end{center}>

Phase 0 gate PASS (□□□). □ Phase 0 ext+ □□□ “SL □□□□□□□” angle □□□ “□□□□□ EVP” □□ (□
Part IV §10 □ docs/spectral_solver_design.md Part V).